Carbon -14 dating

$$^{14}_{6}C \longrightarrow {}^{14}_{7}N + {}^{0}_{-1}\beta + \bar{\nu}$$

So, $A_0 = 0.25\%\ \text{By}\ g^{-1} \times 200g.$

$= 51\ Bg.$ ✓

from $\lambda = \frac{\ln 2}{t_{\frac{1}{2}}}$

$\lambda = \frac{0.693}{5730}$

$= 1.21 \times 10^{-4}\ \text{years}^{-1}.$ ✓

So; $A = A_0 e^{-\lambda t}$

$t = \frac{1}{\lambda} \ln\left(\frac{A_0}{A}\right)$

$= \frac{1}{1.21 \times 10^{-4}} \ln\left(\frac{51}{0.025}\right)$ ✓

$= 5.29 \times 10^{4}$ years old. ✓

Hence the age of the artifact is ≈ 52,900 years old.

Photoelectric effect.

$E = \frac{hc}{\lambda}$ when E is the photon energy for emission to occur $\Longleftrightarrow$ where $E \geq 3.0\,eV$. ✓

for UV radiation; $E = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^{8}}{254 \times 10^{-9}}$

$= 7.79 \times 10^{-19}\ J$

$= 4.87\ eV$

Since $4.87\,eV > 3.0\,eV$, then photoelectric emission occurs which causes the leaf to fall.

for Red light; $E = \frac{6.6 \times 10^{-34} \times 3.0 \times 10^{8}}{600 \times 10^{-9}}$

$= 3.296 \times 10^{-19}\ J$

$= 2.06\ eV$

Since $2.06\,eV < 3.0\,eV$, then photoelectric emission doesn't occurrence there is no change in divergence of the leaf.

Threshold frequency of zinc:

$$f_o = \frac{\phi}{h}$$

$$= \frac{3.0 \times 1.6 \times 10^{-19}}{6.6 \times 10^{-34}}$$

$$= 7.27 \times 10^{14} \text{ Hz}$$

Maximum wavelength or longest wavelength:

from $\phi = \lambda_o f_o$

$$\lambda_o = \frac{c}{f_o}$$

$$= \frac{3.0 \times 10^{8}}{7.27 \times 10^{14}}$$

$$= 4.13 \times 10^{-7} \text{ m}$$

$$= \boxed{413 \text{ nm}}$$

i.e. any radiation with $\lambda < 413$ nm, can cause photoelectric emission from zinc.

Maximum kinetic energy:

$$Ke_{max} = eV_s$$

$$= 1.6 \times 10^{-19} \times 1.2$$

$$= 1.92 \times 10^{-19} \text{ J}$$

or

$$Ke_{max} = \frac{1}{2} mv^2$$

$$= \frac{1}{2} \times 9.11 \times 10^{-31} \times (6.4 \times 10^{5})^2$$

$$= 1.87 \times 10^{-19} \text{ J}$$

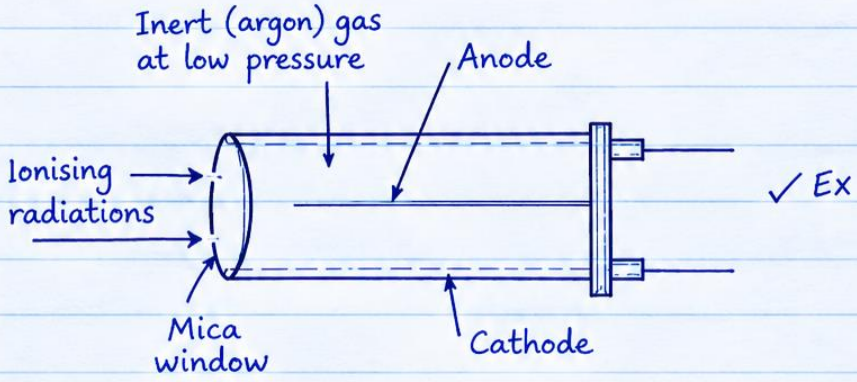
since $\frac{1}{2} mv^2 = eV_s$

$$v^2 = \frac{2eV_s}{m}$$

$$V = \sqrt{\frac{2 \times (1.92 \times 10^{-19})}{9.11 \times 10^{-31}}}$$

$$V = \underline{6.4 \times 10^{5} \text{ ms}^{-1}}$$

(b) The device used to measure radioactivity is a Geiger-Müller tube (GMT)



✓ Ex

Radiations are made incident to the mica window that lets them pass through. This causes ionisation of the gas hence negative ions appear on the anode and positive ions to cathode. Hence a discharge of electrons and a p.d is set up across the circuit and the current is amplified and detected.

Observation from Rutherford's model.

- Some alpha particles pass undiverted ($\theta < 90^\circ$)
- Some particles pass/are deflected at small angle ($\theta < 90^\circ$)
- Some particles are deflected at greater angle ($\theta > 90^\circ$)

~ SCORING GUIDE ~

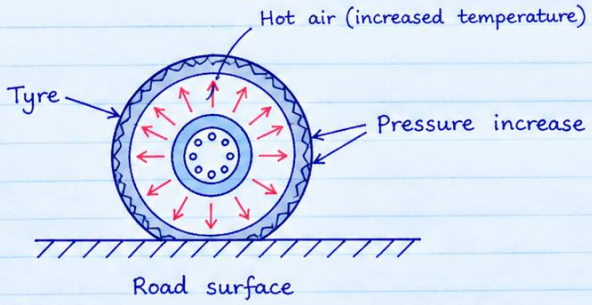
Item 2.

(a) Jerking forward.

Before brakes were applied, both the car and the person made were moving at the same velocity. On applying the brakes suddenly, the force is exerted on the car (not directly on the person), which causes the person to continue moving forward due to inertia. Due to inertia, the person moves forward and is projected towards the front. Hence, the car stops suddenly, but the person continues moving forward.

(b) Car tyre bursting.

It is due to frictional force between the ground and the tyre, which increases the temperature of the air molecules. The molecules move faster, collide with the walls more frequently and more forcefully, hence increasing the pressure inside the tyre.



Force, $F = ma$

$$m = \frac{v - u}{t}$$

$$F = 80\left(\frac{0 - 30}{0.1}\right)$$

$F = -24000$ N ✓ Ans:

$u = 108 \times 1000$

$= 3600$ m/s

$u = 30$ m/s

The retarding force is 24000 N which is the maximum force for minimum injuries. Hence, in turn, the body sustains minimum injuries.

Kinetic Energy, $K.E = \frac{1}{2} mu^2$

$$K.E = \frac{1}{2}(80)(30)^2$$

$K.E = 36000$ J ✓ Ans:

Which is the kinetic energy absorbed by the seat belt for minimum injuries. Hence, the boy sustains them.

(b) Device air bag.

From $F = ma$

$F = m\left(\frac{v - u}{t}\right)$, without air bag, the driver takes a short period of time to hit the steering, thus experiencing a large effect of force. Hence feeling more pain and severe injuries.

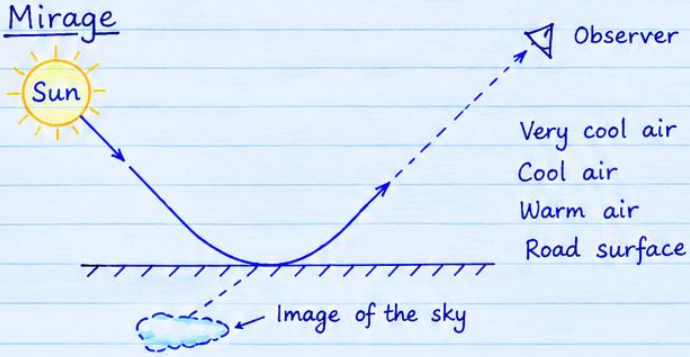
Whereas with air bag, the driver takes a longer period of time to hit the chart on the steering, hence less force of impact, less pain and fewer injuries.



KW	
EX	07
AN	06
AP	05

Item 3

(a) Mirage



On a hot day, the runway (road) heats the road and the road radiates heat upwards toward the sky, warming up the subsequent layers of air in the sky.

The layer of air nearest the road surface is hotter than layers higher up in the sky.

Thus, the density of air relatively increases upwards above the surface of the road.

Light rays from the sky are thus gradually and continuously refracted away from the normal as they pass from denser air to less dense air until at some point the light gets totally internally reflected.

An observer on earth, receiving the totally internally reflected light gets an impression of a pool of water on the road surface, hence a mirage.

(b) Weak and strong signals

The period variation in strong and weak signals is due to interference of radio waves emitted from the telecommunication masts.

The telecommunication masts act as coherent sources emitting radio waves of the same frequency. ✓ EX

At some point, the radio waves from masts arrive out of phase and cancel hence weak signals. ✓ EX

Hence leading to strong and weak signals to result in a clear poor connection. ✓ respectively

Calculation of angle of incidence and critical angle.

(a) Air to glass

$n_g \sin i = n_a \sin\theta$, $\theta = 90° - C$

glass = cladding core

$n_g \sin C = n_a \sin 90°$

$1.60 \sin C = 1.52 \times 1$

$C = \sin^{-1}\left(\frac{1.52}{1.60}\right)$ ✓ (1)

$C = 71.81°$ ✓ (1)

Since $71.81° < 73°$, hence, effective communication.

From $n_a \sin i = n_g \sin 90° = 71.81°$

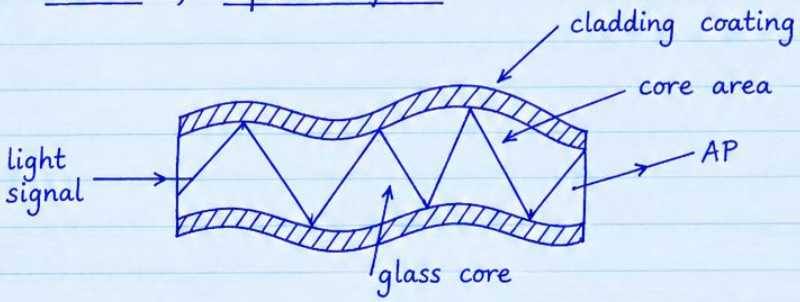
$\sin i = 1.60 \sin 18.19°$

$i = \sin^{-1}(1.60 \sin 18.19°)$

$i = 29.96°$ ✓ (1)

Since $29.96° < 31°$, hence communication is effective. AN✓

(b) Device ; Optical fiber



Optical fiber is made of a core of glass of a higher refractive index that of the cladding core.

Light signal is directed at the boundary between the glass core and the cladding at an angle greater than the critical angle. ✓ AP

The light is totally internally reflected, the same happens inside the point of incidence. AP

Multiple total internal reflections help to transmit light through the fiber without loss of intensity. ✓ AP

Calculation of angle of incidence and critical angle.

This method is accurate because it reduces human errors and improves sensitivity. ✓ EX

(5)

EX	09
AN	04
AP	06
T	19

Item 4

(a) How harmonics are produced.

Harmonics are produced when vibrating object, such as a string, air column, vibrates not only as a whole but also in smaller sections at the same time. ✓ EX

The fundamental frequency (1st harmonic) is produced when the entire length vibrates as one segment. ✓ EX

Higher harmonics are produced when vibrating systems divide into two, three, four or more equal points, creating other stationary wave patterns. ✓ EX (4)

Each harmonic has a frequency that is a whole number multiple of the fundamental frequency i.e. $f_n = nf_o$. ✓ EX

How harmonics affect the quality of sound (timbre).

Timbre is the quality of sound produced. (2)

It depends on the number of harmonics produced by an instrument. ✓ AP

Timbre increases with increase in the number of harmonics. ✓ AP

Calculation for tension in the string.

$V = 2Lf_o$ — ①

for string plucked in the middle; $\lambda = 2L$ — ②

$V = \sqrt{\frac{T}{\mu}}$ — ③

$2Lf_o = \sqrt{\frac{T}{\mu}}$

$\mu = \frac{m}{L} = \frac{4.2 \times 10^{-3}}{0.65} = 0.0065 \text{ kg m}^{-1}$ ✓ AN

$T = \frac{(2 \times 8.8 \times 4 \times 0.65)^2}{0.0065}$

(3)

$T = 741.4 \text{ N}$ ✓ AN

Since $741.4 \text{ N} > 700 \text{ N}$, tension in the string exceeds the set tension; hence increase in tension increases the pitch of the string. ✓ AN

Ice needed for cooling drinks

Mass of ice (M) = 30 − 10 = 20 g = 0.02 kg

$M_{ice,s}\,(30-10) = m_{icp}\,(10-0) + (m_{icw} \times 2.1\times10^{2}) + (m_{ic_s} \times 2.1\times10^{2}\times10)$

$80 \times 4.2 \times 10^{3} \times 20 = (m_i \times 3.26\times10^{3}) + (m_i \times 4.2\times10^{3}\times10)$

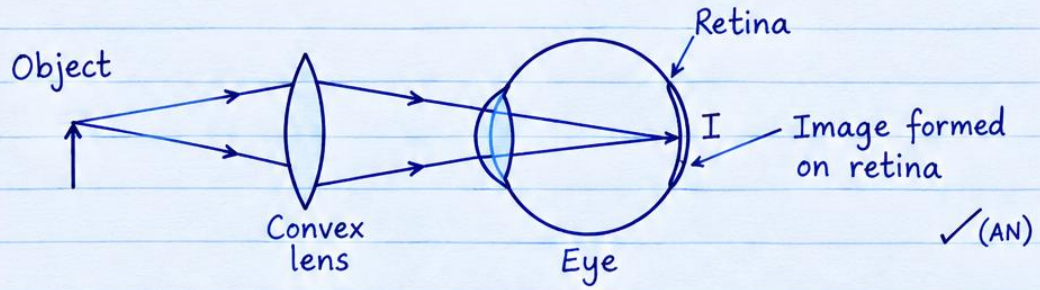
Since 17.8 kg < 2.5 kg, no additional container of ice is needed. ✓(AN)

(b) Grand father is experiencing long sightedness

The eye lens loses elasticity with age; the eye cannot focus nearby objects. Therefore, the images are formed behind the retina, spreading behind it, especially for close objects.

This is corrected by convex lens; causing the light rays to converge before entering the eye, hence the image is formed on the retina behind it. ✓(AN)

Illustration



✓(AN)

Calculation for tension in the string

$V = 2l f_o$ ... ①

for string plucked in the middle; $\lambda = 2l$ ... ②

$V = \sqrt{\frac{T}{\mu}}$ ... ③

$2l f_o = \sqrt{\frac{T}{\mu}}$

$\mu = \frac{m}{l} = \frac{4.2\times10^{3}}{0.65} = 0.065\ \text{kg m}^{-1}$ ✓(AN)

$T = (74.4\ \text{N})^{2} \cdot 0.065 = 745.9\ \text{N}$

Since 741.4 N > 700 N, tension in the string exceeds the set tension; hence increase in tension increases the pitch of the sound. ✓(AN)

Item FIVE (5)

a) Analysis of the battery circuit and atmospheric discharge.

Case I

$$E = I(R + r)$$

$$E = 0.48(21 + r)$$

$$E = 10.08 + 0.48r \quad \checkmark \ (1)$$

Case II

$$E = 0.30(36 + r)$$

$$E = 10.80 + 0.30r \quad \checkmark \ (2)$$

Equating the two equations (1) and (2)

$$10.08 + 0.48r = 10.80 + 0.30r$$

$$0.18r = 0.72$$

$$r = \frac{0.72}{0.18}$$

$$r = 4.0\ \Omega \quad \checkmark \ (AN)$$

Substituting for $r = 4.0\ \Omega$ to obtain E

$$E = 10.08 + 0.30(4)$$

$$E = 10.08 + 1.20$$

$$E = 12.0\ V \quad \checkmark \ (AN)$$

The actual emf is 12 V not 15 V as stated. So the internal resistance r is relatively high at 4.0 Ω. The battery will rapidly lose terminal voltage under load due to its internal resistance which explains why the speaker stops working after a short time.

b) Alternatively:

The battery has an emf of 12.0 V and internal resistance of 4.0 Ω. Since the rated voltage is 15 V, the battery is underperforming, confirming the need for a fault.

Explanation of phenomena

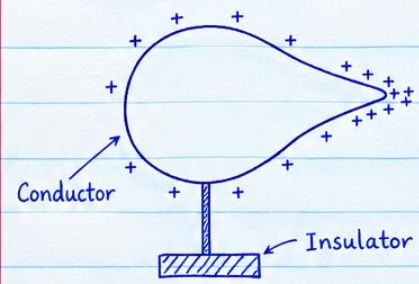
Coronas discharge ✓ EX

The bluish glow and crackling sounds are due to corons discharge. In wet weather, the air around the high-voltage cables becomes ionised ✓EX. Water droplets reduce the dielectric strength of the air, allowing electrons to accelerate and collide with air molecules, causing excitation and emission of light (glow) and ionisation (crackling).

Charge distribution

Charge density is always highest at points with the smallest radius of curvature (r). Or sharp point if ($r \approx 0$). This high concentration creates a strong electric field ✓EX, facilitating the ionisation of surrounding air.

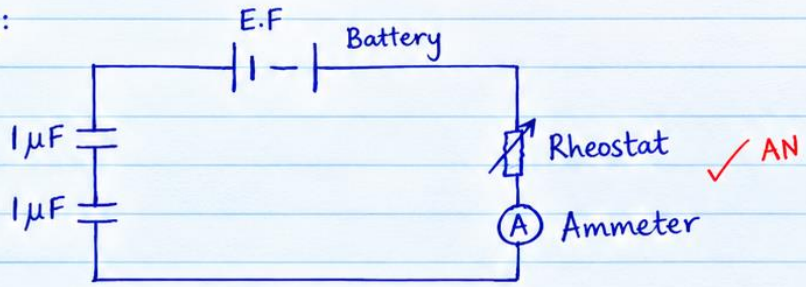
The longest radius of curvature has the lowest charge density.



b) Addressing the warden's concern.

A charger connected to a 240V A.C main socket can charge a 15 DC battery because the charger has a step-down transformer-rectifier unit. This reduces the higher A.C voltage (240V) to a lower A.C voltage suitable for the battery but slightly above 15V', The diode bridge (rectifier) converts that A.C into direct current (D.C) called pulsating D.C. By use of capacitors and filters in the charger the pulsating direct current (D.C) is smoothened. A steady voltage which is then regulated to the required charging voltage for the battery.

Sketch:



✓ AN

From $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$

$\frac{1}{C} = \frac{1}{1} + \frac{1}{1}$

$\frac{1}{C} = 2$

$C = \frac{1}{2} = 0.5\ F$ ✓ AN

For the capacitor to charge, time taken $t = 5(RC)$

So,

$q = 5 \times R \times 0.5$

$R = \frac{q}{2.5}$

$R = \frac{3.4}{2.5} = 1.36\ \Omega$ ✓ AN

$R \approx 1.4\ \Omega$ ✓ AN

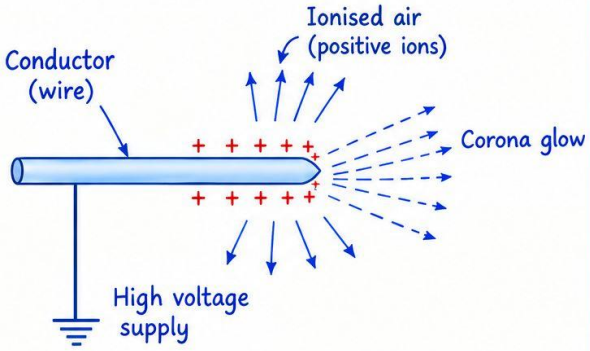


Fig: Corona discharge around a pointed conductor

Summary

1. A high voltage is applied to a conductor, especially at a sharp point (small radius of curvature).
2. The strong electric field at the pointed end ionises the surrounding air, producing positive and negative ions.
3. The free electrons are accelerated towards the conductor, while the positive ions move away from it.
4. The ionised air forms a faint glow around the conductor, called corona glow, and a small current flows through the air.

Resonant tuning circuit in radios.

It occurs when $X_L = X_C$; $f = f_o$

$$f_o = \frac{1}{2\pi f_o C}$$

$$f_o = \frac{1}{2\pi\sqrt{LC}}$$

$$f_o = \frac{1}{2\pi\sqrt{0.5\times10^{-6}\times10}}$$ ✓AN

$f_o = 159\,Hz$; hence the station is captured. ✓AN

Current flowing in the wire.

$F = BIL$

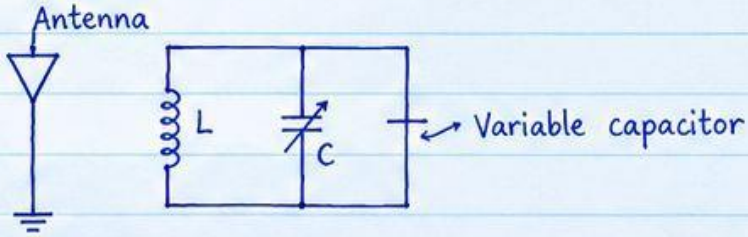
$mg = BIL$

$$I = \frac{0.20\times10^{-3}\times10)}{0.20\times5\times10^{-2}}$$ ✓AN

$I = 2.0A$ ✓AN

Since $2.0A > 1.9A$, current flowing through the wire is maximum. At maximum current, $X_L = X_C$, then the opposition to the flow of current is minimum. Hence the radio circuit can easily select the broadcasting frequency clearly. ✓EX

(b) Device is a radio receiver. ✓AP



This works by receiving electromagnetic waves through antenna. AP
These radio waves induce alternating electrical signals in the antenna by electromagnetic induction. ✓AP
The LCR circuit then selects the desired frequency at $X_L = X_C$ at maximum current flow ✓AP
Therefore one station is clearly received and electrical signals are converted into sound by the loudspeaker in the radio. ✓AP

Since $R = 4.0\,\Omega$, this confirms the battery's internal resistance, and the charging time of about 9s as determined by the task.

The battery's internal resistance reduces the charging current, which in turn reduces the charging time. Hence the time is about 9s.

Therefore the required answers are:

$$q = 5 \times R \times 0.5$$

$$R = \frac{9}{2.5}$$

$$R = 3.4\,\Omega$$

$$R \approx 4.0\,\Omega \quad \checkmark AN$$